HeatWise — User Flow Document

Version: 2.1 | **Date:** March 2026 | **Status:** Production-ready demo

1. Product Overview

HeatWise is an AI-powered urban rooftop greening platform for Indian cities. It helps homeowners, housing societies, and commercial property managers:

- Measure their rooftop or balcony using AR/camera
- Receive an AI-generated green installation plan (plants + layout)
- Understand the predicted cooling, CO_2, and water savings
- Request a quote from certified local installers

2. User Personas

Persona	Description	Primary Goal
Urban Homeowner	Flat/villa owner, 28–55 yrs, concerned about heat	Reduce cooling bills, make roof usable
Housing Society Mgr	Manages 50–500 unit complex	Large-scale green roof ROI
Commercial FM	Facility manager for office/retail	Compliance, ESG reporting
Installer Partner	Certified green-roof contractor	Receive qualified leads and quotes

3. End-to-End User Flow

```
┌──────────────────────────────────────────────────────────────┐
│                    HEATWISE USER JOURNEY                     │
└──────────────────────────────────────────────────────────────┘


  [Download / Open App]
         │
         ▼
  ┌──────────────┐
  │   SPLASH     │  Boot animation + brand intro (2.5s)
  └──────────────┘
         │
         ▼
  ┌──────────────┐
  │  ONBOARDING  │  3-slide carousel: What HeatWise does
  │   SLIDES     │  (skippable on 2nd+ launch)
  └──────────────┘
         │
         ▼
  ┌──────────────┐
  │  PHONE AUTH  │  Enter mobile number → OTP sent via SMS
  │  (Sign In)   │  ──► Dev mode: OTP shown on screen
  └──────────────┘
         │
         ▼
  ┌──────────────┐
  │   PROFILE    │  Name, city, gardening interest (1-time)
  │  COMPLETE    │  Skippable, can finish later
  └──────────────┘
         │
         ▼
  ┌──────────────────────────────────────────────────────────┐
  │                     HOME DASHBOARD                       │
  │  • City Cooling Index score (animated)                   │
  │  • Quick metrics: area scanned, temp drop, CO₂, energy   │
  │  • Recent projects list (from DB)                        │
  │  • CTA: "Start New Scan"                                 │
  └──────────────────────────────────────────────────────────┘
         │
         │  [Start New Scan]
         ▼
  ┌──────────────┐
  │   SCAN /     │  Choose scan method:
  │  MEASURE     │  A) Live AR measurement (WebGL camera)
  │   SCREEN     │  B) Manual input (length × width)
  │              │  Calculates area in m² in real time
  └──────────────┘
         │
         ▼
  ┌──────────────┐
  │  PROJECT     │  Name, location, surface type, primary goal
  │  DETAILS     │  Budget range, floor level, load capacity
  │   FORM       │  Space type: balcony / terrace / rooftop
  └──────────────┘
         │
         ▼
  ┌──────────────┐
  │ ENVIRONMENT  │  Sun exposure, water availability, wind level
```

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│  SETTINGS   │  Region (auto-populated via geospatial API)
│             │  Maintenance preference, special needs (pet-safe)
└─────────────┘
       │
       ▼
┌─────────────┐
│  AI ANALYSIS│  POST /api/recommendations/generate
│  (loading)  │  • ML model scoring (Python serving)
│  NeuralBrain│  • Catalog hybrid fallback
│   3D anim   │  • Rules-only fallback (always available)
└─────────────┘
       │
       ▼
┌──────────────────────────────────────────────────────────┐
│                      RESULT SCREEN                       │
│  • 3D Garden Model (Three.js) — species placed on roof   │
│  • Metrics: Green Area / Water/Day / Species / Energy saving  │
│  • Filterable species cards (Sun / Wind / Cooling filters)    │
│  • CO₂ projection chart (3-year)                         │
│  • Confidence level from ML model                        │
│  CTAs:  [Generate AI Visual] [Request Quote] [Full Report]   │
└──────────────────────────────────────────────────────────┘
       │
       │
       ├──────────────────────────┐
       │                          │
       ▼                          ▼
┌─────────────┐            ┌──────────────┐
│  BEFORE /   │            │  FULL REPORT │
│   AFTER     │            │   SCREEN     │
│ AI VISUAL   │            │ Tabs: THERMAL │
│ (OpenAI     │            │ SPECIES IMPACT│
│  Vision)    │            │              │
└─────────────┘            └──────────────┘
       │                          │
       └──────────────────────────┘
                  │
                  ▼
┌──────────────────────────────────────────────────────────┐
│              INSTALLATION QUOTE REQUEST                  │
│  • Project scope summary (area, cooling, confidence)     │
│  • Optimal install window (dynamic — next 2 months)      │
│  • Contact details form (name, email, phone)              │
│  • Timeline preference                                    │
│  • Partner network selection (local certified installers)  │
│  • POST /api/installers/quote-request                     │
└──────────────────────────────────────────────────────────┘
       │
       ▼
┌─────────────┐
│  SUCCESS /  │  Confirmation + installer contact promise
│  DASHBOARD  │  Return to Home → project saved in DB
└─────────────┘
```

4. Screen-by-Screen Summary

4.1 Splash Screen

- Duration: ~2.5 seconds
- 3D MegaCity animation in background
- HeatWise brand + version
- Auto-advances to Onboarding (first launch) or Home (returning user)

4.2 Onboarding (3 slides)

- **Urban Heat** — Problem statement, city temperature data
- **Smart Greening** — How AI + AR solves it
- **Measurable Results** — Cooling, CO_2, water metrics

4.3 Phone Auth

- Enter `+91 XXXXXXXXXX` format
- OTP delivered via SMS (dev: displayed on screen)
- 6-digit code entry screen
- Creates User record in DB on first sign-in

4.4 Complete Profile

- Name, city selection
- Gardening interest level (1–5)
- One-time screen, skippable

4.5 Home Dashboard

- City Cooling Index (0–100, animated count-up)
- 4 key metrics tiles
- Recent project list from `/api/projects`
- Bottom nav: Home / Scan / Report / Settings

4.6 Measure Screen

- AR camera mode (Live AR with Three.js overlay)
- Manual mode: L × W inputs → real-time area
- Output: `widthM`, `lengthM` stored in `photoSession`

4.7 Environment Screen

- Sun exposure: Full / Partial / Shade
- Water availability: Abundant / Moderate / Scarce
- Wind level: Low / Medium / High

- Maintenance preference: Minimal / Low / Moderate / High

4.8 Analysis Screen

- NeuralBrain 3D visualization while processing
- Calls `POST /api/recommendations/generate`
- Shows progress steps: Scanning → Processing → Generating

4.9 Result Screen

- **Hero**: 3D garden model with labeled plant species
- **Metrics**: Green Area, Water/Day, Species Count, Energy Saving
- **Species cards**: Filterable by sun / wind / cooling score
- **CO_2 chart**: Year 1 → Year 3 projection
- **Action buttons**: AI Visual / Install Quote / Full Report

4.10 Report Screen

Three tabs:

- **Thermal** — Temperature reduction chart, before/after bars
- **Species** — All recommended plants with descriptions, water needs, pet-safety
- **Impact** — CO_2 offset, water conservation, biodiversity score, heat reduction

4.11 Install Screen

- Dynamic optimal install window
- Contact form
- Installer network RFQ
- On submit: creates `InstallerQuoteRequest` + `InstallerInstallJob` in DB

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5. Navigation Map

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                    ┌──────────┐
                    │  SPLASH  │
                    └──────────┘
                         │
                    ┌────▼─────┐
                    │ ONBOARDING │
                    └──────────┘
                         │
            ┌────────────▼────────────┐
            │       PHONE AUTH        │
            │  send-otp → verify-otp  │
            └─────────────────────────┘
                         │
            ┌────────────▼────────────┐
            │     COMPLETE PROFILE    │
            └─────────────────────────┘
                         │
  ┌──────────────────────▼──────────────────────┐
  │                HOME DASHBOARD                │
  │            (bottom nav: home tab)            │
  └──────────────────────────────────────────────┘
     │                 │                 │
[New Scan]        [My Scans tab]    [Settings tab]
     │
     ▼
┌─────────┐    ┌──────────┐    ┌──────────┐
│ MEASURE │──→ │ ENV SETUP │──→ │ ANALYSIS │
└─────────┘    └──────────┘    └──────────┘
                                     │
                                ┌────▼────┐
                                │ RESULT  │
                                └─────────┘
                                  │ │ │
                          ┌───────┘ │ └──────────┐
                          ▼         ▼            ▼
                      ┌────────┐ ┌────────┐ ┌──────────┐
                      │BEFORE/ │ │ REPORT │ │ INSTALL  │
                      │ AFTER  │ │        │ │  QUOTE   │
                      └────────┘ └────────┘ └──────────┘
```

6. Data Flow Diagram

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USER INPUT                  APP LAYER                  BACKEND / DB
__________                  _________                  ____________


Phone Number ──────────► /api/phone-auth/send-otp ──► PhoneOtp (DB)
OTP Code ──────────────► /api/phone-auth/verify-otp ─► User (DB)


Scan dimensions ─────► photoSession (React state)
Environment prefs ───► buildRecommendationGenerateRequest()
                            |
                            ▼
                  /api/recommendations/generate
                            |
                  ┌─────────┼──────────────────┐
                  ▼         ▼                  ▼
            Python ML    Catalog Hybrid       Rules-Only
            Serving      Fallback (TS)        Fallback (TS)
                  |         |
                  └─────────┘
                            |
                            ▼
                  ScoredPlants[] + HeatReductionSummary
                            |
                            ▼
                  RecommendationTelemetrySession (DB)
                  RecommendationCandidateSnapshot (DB)
                            |
                            ▼
                      React: ResultScreen
                            |
                  ┌─────────┼──────────┐
                  ▼                    ▼
            /api/reports/        /api/installers/
            recommendation-      quote-request
            dossier              InstallerQuoteRequest (DB)
                  |
                  ▼
            /api/reports/user-report
            RecommendationDossier (DB)
```

7. Key User Actions & API Calls

User Action	API Endpoint	DB Tables Written
Request OTP	`POST /api/phone-auth/send-otp`	`PhoneOtp`
Verify OTP	`POST /api/phone-auth/verify-otp`	`User`
Load projects	`GET /api/projects`	—

Generate recommendation	`POST /api/recommendations/generate`	`RecommendationRun`, `RecommendationTelemetrySession`
View results	— (client-side)	`RecommendationTelemetryEvent` (view)
Generate AI visual	`POST /api/generate-garden-visual`	— (OpenAI call)
Request quote	`POST /api/installers/quote-request`	`InstallerQuoteRequest`
View full report	`POST /api/reports/recommendation-dossier`	`RecommendationDossier`

8. Error States & Fallbacks

Scenario	Fallback Behaviour
ML Python process unavailable	Catalog hybrid TypeScript fallback
Catalog hybrid fails	Rules-only TypeScript fallback
No DB connection	Auth fails gracefully with error message
No internet / API offline	"Load failed" with retry button
Zero species match filters	Show full unfiltered species list
No photo captured	Skip AI visual option (button hidden)

Document maintained by HeatWise engineering. Update after each sprint.